Simulating Equity: An Agent-Based Audit of Effective Accessibility for the MARTA Transit System

Brief Project Description

This project utilizes **Agent-Based Modeling (ABM)** to evaluate the gap between theoretical and effective transit access in Atlanta. While traditional planning tools measure accessibility based on distance to a MARTA stop, this study introduces a human-centered simulation that accounts for real-world friction such as sidewalk quality, intersection safety, and bus service reliability (using **GTFS-Realtime data**). By simulating agents with varying mobility constraints navigating the last-mile environment, the project identifies where the municipal economy of transit fails to serve its most vulnerable users. The goal is to provide a critical, data-driven framework for prioritizing infrastructure improvements that ensure transit functions as a true public good.

Section 2: Data and Methodological Approach (250–300 Words)

The methodology for this research employs a hybrid approach, integrating macro-level accessibility analysis with micro-level **Agent-Based Modeling (ABM)** to quantify Effective Accessibility. The study will be focused on a high-transit-dependent corridor in Atlanta (e.g., the Buford Highway or Westside corridors).

Data Sources: The project will utilize three primary data streams:

1. **Static Spatial Data:** I will aggregate sidewalk inventories, street lighting locations, and crosswalk data from the **Atlanta Regional Commission (ARC)** and OpenStreetMap (OSM) to build a friction layer representing physical walking conditions.
2. **Dynamic Transit Data:** I will utilize **GTFS-Realtime** feeds from MARTA to analyze service reliability and headway variances, moving beyond static schedules.
3. **Social Vulnerability Data:** Census-tract level data for rent-to-income ratios and eviction rates will be used to identify areas of high precarity.

Methodological Approach: The research begins with a **Multi-Criteria Decision Analysis (MCDA)** in ArcGIS Pro to establish a baseline of Potential Access based on distance and service frequency. I will then develop a **Python-based ABM** (utilizing the Mesa or GeoPandas libraries) to stress-test this baseline. In the simulation, agents with varying mobility constraints such as walking speed variations or strict time-windows for work will navigate the network. Unlike traditional models, these agents will encounter friction from missing infrastructure and bus delays. By comparing the success rate of agents reaching essential destinations (clinics, grocery stores, jobs) in the simulation versus the theoretical coverage in official maps, I will quantify the accessibility gap. This approach allows me to move from abstract ridership metrics to a community-centered analysis of how the municipal budget manifested in infrastructure quality facilitates or hinders economic mobility.

Section 3: Learning Objectives and Expectations (250–300 Words)

Completing this capstone is a pivotal step in my transition from a graduate student in **Urban Analytics** to a professional transportation planner. My expectations for this project are divided into technical mastery, theoretical grounding, and professional application.

Technical Mastery: While I have experience with GIS and Python, this project requires a significant leap in **data engineering and simulation logic**. I expect to learn how to integrate dynamic GTFS-Realtime feeds into a spatial model, a skill that is increasingly in demand for smart-city planning. Furthermore, refining my Agent-Based Modeling skills will allow me to transition from academic simulations (like my Tokyo pedestrian project) to a rigorous audit tool that can influence real-world policy decisions in Atlanta.

Theoretical Grounding: Working under Professor Barrett, I expect to deepen my understanding of **Critical Urban Theory**. I want to learn how to translate qualitative concepts such as social resilience and emancipatory politics into quantitative spatial weights. This capstone will teach me how to critically evaluate regimes of data, identifying whose mobility is being prioritized and whose is being ignored by standard planning software. Learning to name these gaps with data is essential for my goal of being an advocate for equity-based planning.

Professional Application:. By framing transportation as a public good that requires protective social infrastructure, I hope to demonstrate how Urban Analytics can move beyond efficiency toward justice. This project will serve as my primary professional portfolio piece, proving that I can manage a complex, independent research lifecycle from data scraping and model calibration to final policy recommendations which is fundamental for my career path in regional transportation planning.